Shennong's Glass: training-free foundation-model phenotypic screening with calibrated uncertainty for organ-on-chip drug evaluation

Shreshth Mishra— AI4S Open Innovation: AI for Life Science

Category: **End-to-End System** · Code: https://github.com/shreshth006/shennongs-glass

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Abstract

Organ-on-chip (OoC) and other microphysiological systems produce rich imaging read-outs, but turning those images into decisions — *is this compound active, how does it act, at what dose does it become toxic, and can I trust the call?* — still depends on hand-tuned image analysis and expert inspection. We present **Shennong's Glass**, an end-to-end system that takes standard three-channel endpoint stains (nuclei, β-tubulin, F-actin) and returns (i) a phenotypic activity score calibrated against vehicle controls, (ii) a viability read-out and Hill-fitted dose–response, (iii) a mechanism-of-action (MoA) ranking with a *conformal prediction set* of guaranteed coverage, (iv) a novel-mechanism alarm and (v) a plain-language explanation of what changed in the cells. The image encoder is a frozen self-supervised foundation model (DINOv2), so no task-specific training or labels are needed and the system transfers to a new chip by swapping its reference library. On the BBBC021 benchmark (5,775 fields, 113 compounds, 8 doses each) Shennong's Glass reaches 85.4% MoA accuracy on held-out compounds (NSC) and 77.2% when also holding out imaging batches (NSCB); its 90% conformal sets achieve 91.3% empirical coverage with 1.18 classes on average, and it flags mechanisms absent from its library with AUROC 0.79. Across 112 compounds it separates phenotypic potency from cytotoxicity and triages compounds by the width of that window. Everything — features, figures and the interactive demo — is reproducible with free Kaggle notebooks and a laptop CPU.

Shennong, the Divine Farmer of Chinese legend, tasted hundreds of herbs to tell medicine from poison; his body, the story goes, was transparent, so that he could watch each herb act on his organs. An organ-on-chip is that transparent body. Shennong's Glass is the eye that watches it.

1 Problem and application scenario

OoC platforms recapitulate tissue-level physiology and are increasingly used for efficacy and toxicity testing before animal or human studies [8, 9]. Their most common quantitative endpoint is still fluorescence imaging of fixed tissue: nuclei (DAPI/Hoechst) for cell number and nuclear state, and cytoskeletal stains — β-tubulin and F-actin — for cell shape, neurite/axon integrity and barrier morphology. Neural OoC platforms such as those of the challenge supporter rely on exactly these markers, since microtubule integrity is what chemotherapy-induced peripheral neuropathy destroys [10].

Three practical gaps limit how much value is extracted from such images:

1. **Analysis is assay-specific.** Pipelines are tuned per cell type and per question, and deep models trained on one assay rarely transfer; OoC datasets are too small to train them.
2. **Calls come without reliability.** A classifier always outputs *some* MoA, even for an inactive compound or a mechanism it has never seen; a screening scientist needs to know when not to trust it.
3. **Potency and toxicity are judged separately.** A morphological effect is only interesting if it appears at doses below overt cytotoxicity.

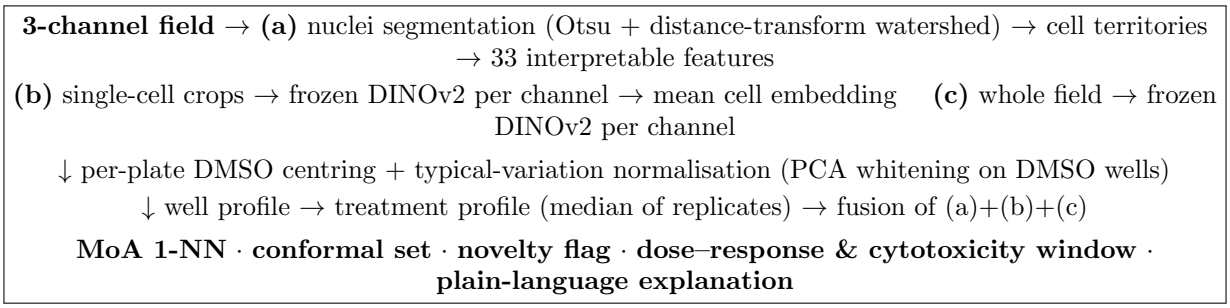


Figure 1: Shennong's Glass pipeline. No component is trained on BBBC021 labels.

Shennong's Glass addresses all three in one system, and presents the result in an interface a biologist can use without writing code (Section 6).

2 Data, licences and compliance

BBBC021v1 [1, 2]: MCF-7 breast-cancer cells treated for 24 h with 113 small molecules at 8 concentrations (plus DMSO vehicle and positive controls), imaged in three channels (DAPI, β-tubulin, F-actin) at 4 fields per well on 55 plates in 10 weekly batches; 13,200 fields in total. A curated subset of 103 compound–concentration *treatments* (38 compounds) is annotated with one of 12 MoAs. We used fields of view 1 and 3 of every well (5,775 fields, 3,300 wells; Section 7). Images and labels are copyright AstraZeneca and distributed for research by the Broad Bioimage Benchmark Collection; we do not redistribute them — our scripts fetch them from the public sources. The dataset contains no human or patient data.

Models and libraries. DINOv2 ViT [5] (Apache-2.0), PyTorch, scikit-image, scikit-learn, UMAP, Gradio (all open-source licences). No commercial API is used.

3 Method

3.1 Three complementary views of each field

(a) Interpretable morphology. Nuclei are segmented on the DAPI channel at half resolution (Gaussian smoothing, Otsu threshold, distance-transform watershed); each nucleus is grown by up to 20 px into the cytoskeletal foreground to approximate its cell. Per cell we measure nuclear area, perimeter, eccentricity, solidity, axis lengths, DNA intensity and integrated DNA content, cell area, tubulin and actin intensities, cytoplasm-to-nucleus ratio and cell elongation, and per field the cell count, channel coverage and edge-texture of each channel. Fields are summarised by the median and inter-quartile range across cells (33 features). These features carry the explanations shown to users and double as a classical baseline.

(b) Single-cell foundation-model embedding. Around a random subset of segmented nuclei we cut fixed-size crops (one cell plus its immediate neighbourhood). Each channel is embedded separately by a frozen DINOv2 ViT (grey-scale replicated to RGB, ImageNet normalisation) and the class tokens are averaged over cells, then concatenated across channels. Embedding channels separately avoids imposing a false RGB semantics on fluorescence stains and keeps the method applicable to any number of channels.

(c) Whole-field embedding. The full field is embedded per channel (class token and mean patch token), capturing tissue-level context — density, clustering, debris — that single-cell crops miss.

3.2 Normalisation anchored on vehicle controls

Plate and batch effects dominate raw image features. For every plate we subtract the mean of its DMSO wells; we then fit a PCA on DMSO wells only and whiten (typical variation normalisation, TVN [4]), so that the remaining axes describe drug-induced rather than nuisance variation. Classical features are robust-z-scored against the plate's DMSO median and pooled DMSO MAD. Fields are averaged into wells and wells into treatments (median over replicates). The fusion profile concatenates the ℓ_2-normalised blocks (b), (c) and (a) (weight 0.7).

3.3 Mechanism of action and calibrated uncertainty

MoA ranking. A query profile is compared by cosine similarity to the reference treatments; the score of a MoA is the similarity to its closest reference, turned into probabilities with a temperature softmax. This nearest-neighbour design needs no training, makes every call traceable to concrete reference compounds (shown in the app) and lets users add new references instantly.

Conformal prediction sets. Instead of a single label Shennong's Glass reports the smallest set of MoAs that contains the truth with user-chosen probability $1 - \alpha$. We use split-conformal prediction [6, 7] with non-conformity $1 - \hat{p}(\text{true MoA})$ calibrated *leave-one-compound-out*: for each query compound the threshold is computed on all other compounds only. Under exchangeability of compounds this guarantees marginal coverage $\geq 1 - \alpha$ on new compounds; large sets honestly signal ambiguity.

Novel-mechanism detection. If the maximum similarity to the library falls below the 5th percentile observed for in-library compounds, the call is flagged as a possible new mechanism. We evaluate this by hiding each query's own MoA from the library.

3.4 Dose–response, cytotoxicity and triage

Viability of a well is its cell count relative to the median DMSO well on the same plate. **Phenotypic activity** is the distance of the well's single-cell profile from the DMSO centre in whitened space; its null distribution is given by the DMSO wells themselves, and a well is called active above the 99th percentile of that null (1% false-positive rate on vehicle); a compound is active from the lowest dose at which this holds for that dose *and* the next higher one, which controls false positives across the eight-dose series. Both read-outs are fitted with four-parameter Hill curves on $\log_{10}$ concentration. From the dose series we derive the lowest active and lowest cytotoxic (viability $< 50\%$) concentrations, their ratio — the *phenotype-to-cytotoxicity window* — and a triage label: inactive; cytotoxic without a distinct phenotype; active and non-cytotoxic; active $\geq 10\times$ below cytotoxicity; narrow window; phenotype only at cytotoxic doses.

3.5 Explanations

For every MoA we report its median morphological signature (Fig. 7); for every query, the classical features deviating most from DMSO are rendered as sentences (e.g. "nuclear area increased (+3.1 SD vs. DMSO)").

4 Implementation

Feature extraction (`kaggle/features/features.py`) runs as Kaggle notebooks over the public BBBC021 mirror; analysis (`scripts/run_analysis.py`, < 1 min on a laptop CPU) produces all numbers and figures in this report from the extracted features; numbers in the text are inserted automatically from `results/metrics.json`. The demo (`app/app.py`, Gradio) reuses the identical extraction code (`shennong/extract.py`) on uploaded images. The configuration used for the reported results is a DINOv2 ViT-S/14, 16 single-cell crops per field (128×128 px, embedded at 112×112) and fields embedded at 224×280; the extraction script also supports ViT-B/14 at higher resolution for GPU runs.

5 Experiments and results

5.1 Mechanism-of-action benchmark

We follow the standard protocol [3]: 1-nearest-neighbour MoA assignment of the 103 annotated treatments, never matching a treatment to another dose of the same compound (*not-same-compound*, NSC). The stricter NSCB additionally forbids matches to treatments imaged in the same batch and, as is customary, drops the two MoAs that occur in a single batch (92 treatments).

Profile	NSC	NSCB
Classical morphology (33 interpretable features)	70.9%	59.8%
DINOv2 single cells, plate-centred only (no TVN)	79.6%	67.4%
DINOv2 whole field + TVN	79.6%	71.7%
DINOv2 single cells + TVN	85.4%	80.4%
Fusion (Shennong's Glass)	**85.4%**	**77.2%**

Table 1: 1-NN MoA accuracy on held-out compounds (main configuration). For context, the best published pipelines on this benchmark — hand-engineered CellProfiler profiles with factor analysis, or deep networks trained on BBBC021 — report roughly 90–96% NSC [3, 4]. Our training-free, reduced-compute configuration is below that range; its value lies in needing no labelled training data, in its calibrated uncertainty (Section 5.2) and in its stability across design choices (Table 2).

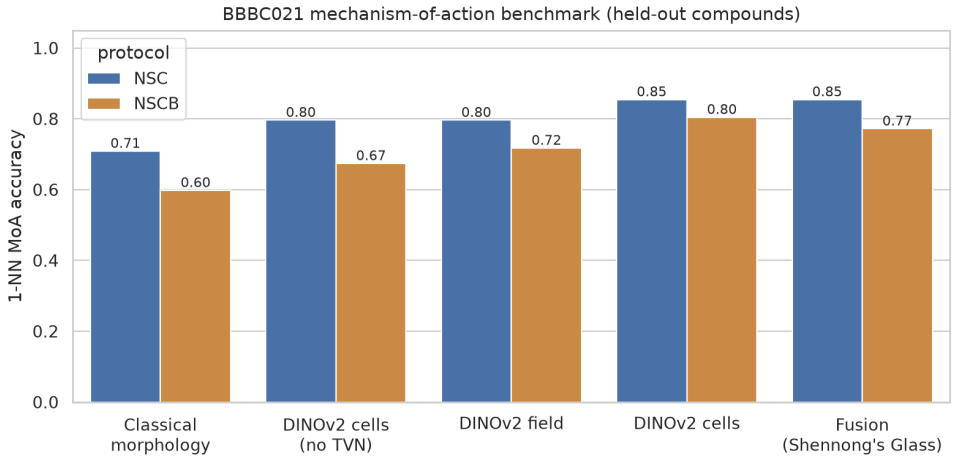


Figure 2: MoA accuracy by profile type and protocol.

Robustness to design choices. The main configuration (128 TVN components, classical-block weight 0.7) was fixed in code before any result was computed. Because 103 treatments make each percentage point roughly one treatment, Table 2 reports the benchmark across the alternatives we considered rather than the best one. Single views move substantially with the number of whitening components (78.6%–90.3% NSC over the whole grid), whereas the fused profile stays within 84.5%–88.3% NSC in every setting: combining views mainly buys *stability*. Full-rank whitening (330 components, one per DMSO well) was also tested and is worse (82.5% NSC), as expected when the vehicle covariance is estimated from as many wells as dimensions.

Profile	TVN 32	TVN 64	TVN 128	TVN 256
cells	82.5 / 70.7	79.6 / 73.9	85.4 / 80.4	90.3 / 83.7
field	84.5 / 79.3	81.6 / 79.3	79.6 / 71.7	78.6 / 73.9
cells+field	87.4 / 81.5	86.4 / 80.4	82.5 / 79.3	89.3 / 81.5
fusion (w=0.4)	88.3 / 79.3	86.4 / 78.3	84.5 / 79.3	88.3 / 83.7
fusion (w=0.7) (main)	88.3 / 77.2	84.5 / 72.8	85.4 / 77.2	87.4 / 80.4

Table 2: NSC / NSCB (%) across numbers of TVN components and fusion weights. Median over the grid: 85.4% NSC, 79.3% NSCB.

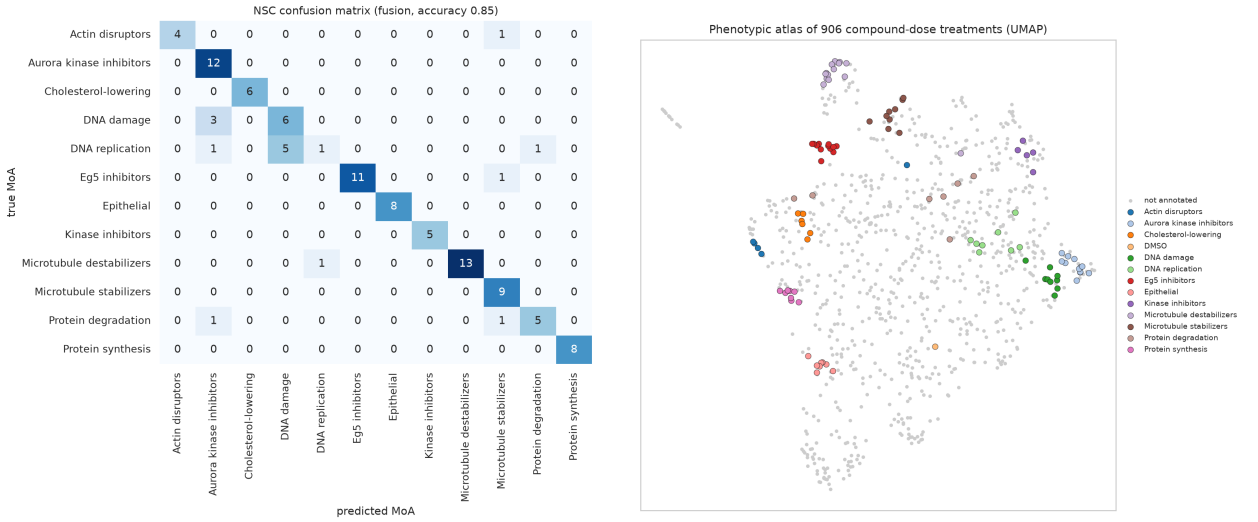


Figure 3: Left: NSC confusion matrix of the fusion profile. Right: UMAP of all 906 compound–dose treatments; grey points are the ∼800 un-annotated treatments, which Shennong's Glass also places and labels.

Mechanisms the library was never told about. About 800 of the 906 treatments carry no MoA label. Among them, *vinblastine* — a vinca alkaloid and a classic cause of chemotherapy-induced peripheral neuropathy [10] — is called a microtubule destabiliser at 7 of its 8 doses, from 3 nM, with probabilities 0.71–0.98; the unannotated tubulin binder *podophyllotoxin* receives the same call at its active dose. Calls made only at overtly cytotoxic doses are less specific (e.g. camptothecin and mitomycin C are labelled microtubule-like at their top doses), which is why the triage reports the cytotoxicity window alongside every mechanism call.

5.2 Reliability

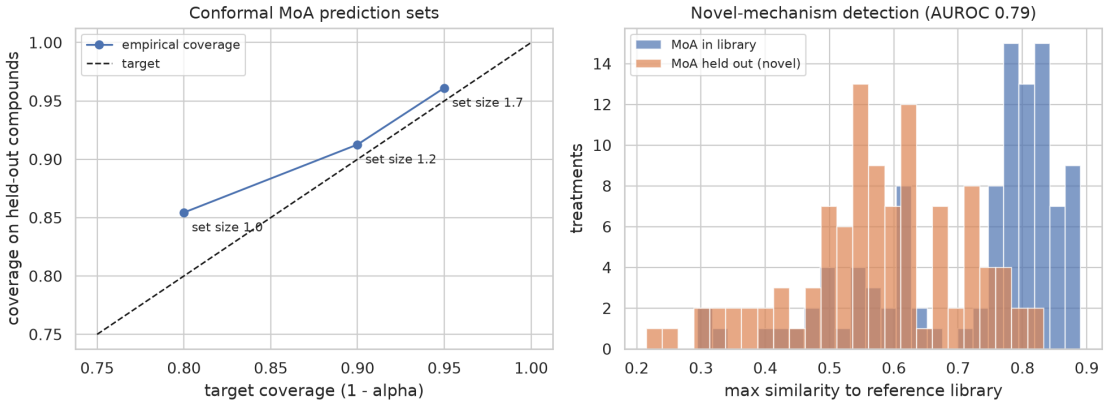


Figure 4: Left: conformal coverage on held-out compounds versus target. Right: maximum similarity to the library when the true MoA is present (blue) or hidden (orange).

Conformal sets meet their target: 96.1% coverage at 95% (mean set size 1.73), 91.3% at 90% (size 1.18; 83.5% of sets are single MoAs) and 85.4% at 80% (size 1.00). Hiding a query's own mechanism lowers its best library similarity enough to be detected with AUROC 0.79. Finally, 88.3% of the annotated (active) treatments have replicate wells more similar to each other than the 95th percentile of random well pairs (*percent replicating*); across all 906 treatments, including the many at sub-active doses that are not expected to replicate, the figure is 37.3%.

5.3 Dose–response and cytotoxicity window

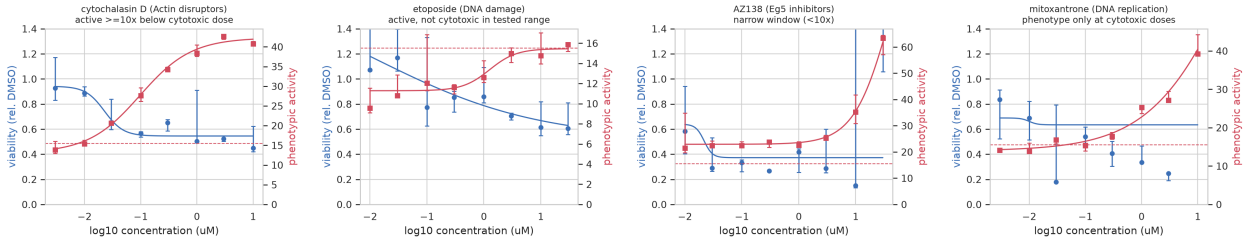


Figure 5: Viability (blue) and phenotypic activity (red; median and inter-quartile range over wells; dashed line = 99th percentile of DMSO) with Hill fits, for one annotated compound of each triage class (selected automatically: the most active compound of the class).

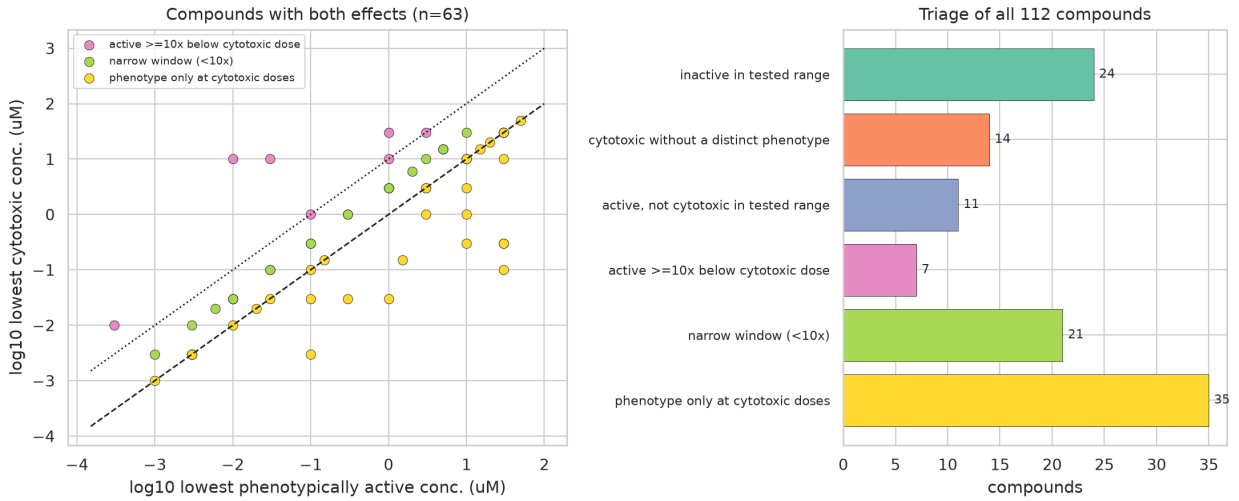


Figure 6: Left: lowest phenotypically active versus lowest cytotoxic concentration for the compounds that show both (dashed: equal; dotted: 10× window). Right: triage of all compounds.

Of 112 compounds, 24 show neither a phenotype nor cytotoxicity in the tested range and 14 reduce cell number without a sustained specific phenotype; 11 are active without overt cytotoxicity, 7 act at least 10× below their cytotoxic dose, 21 have a narrower window and 35 only change morphology at doses that already reduce viability. This separation is exactly the information needed to prioritise compounds — or, for a toxicity panel, to flag those whose cytoskeletal phenotype precedes cell loss.

5.4 Interpretability

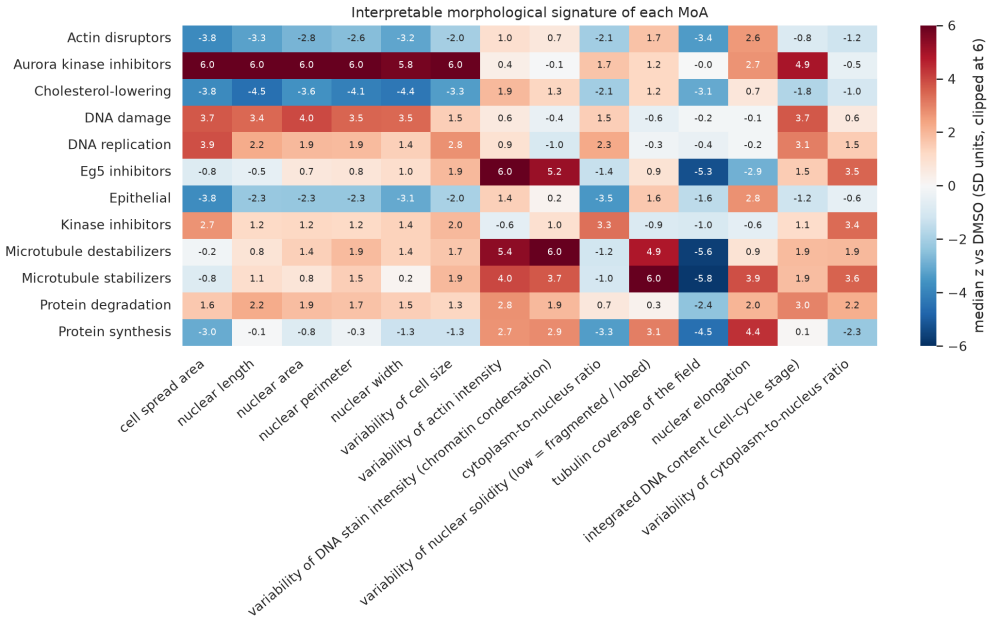


Figure 7: Median morphological signature of each MoA (z vs. DMSO in SD units; the 14 most MoA-discriminative features). The signatures recover known biology without supervision: Aurora-kinase inhibitors produce giant, DNA-rich nuclei (polyploidy after failed cytokinesis); DNA-damaging and replication inhibitors enlarge nuclei and raise DNA content (S/G2 arrest); Eg5 inhibitors and microtubule stabilisers/destabilisers increase the heterogeneity of DNA and tubulin staining while reducing tubulin coverage (rounded mitotic cells with condensed chromosomes); protein-synthesis inhibitors, actin disruptors and statins shrink cell spread area.

6 Interactive system

The Gradio application has four tabs. **Analyze images**: upload the three channels of a field; Shennong's Glass shows the segmentation overlay, the activity score with its percentile among vehicle fields (active / borderline / inactive), the MoA ranking with 90% conformal set when the field is active, a novelty warning when appropriate, the morphological changes in words, and the six most similar reference treatments with thumbnails. A *blind-test* switch removes a chosen compound from the library, so the bundled examples are evaluated honestly. **Compound explorer**: dose–response, cytotoxicity window, dose-wise MoA calls (suppressed at inactive doses) and representative images for every compound. **Phenotypic atlas**: interactive UMAP of all 906 treatments. **Benchmark**: the validation results above.

7 Reliability analysis and limitations

- **Not yet OoC data.** BBBC021 is a 2D monolayer of one cancer line. The pipeline only assumes the three stains, but 3D perfused chips add z-stacks, mixed cell types, PDMS autofluorescence and channel geometry. The next step is to rebuild the reference library on chip images (e.g. neural chips treated with reference neurotoxicants) — no retraining is required, but validation is.
- **Reduced sampling.** Kaggle GPU sessions were unavailable while this work was done, so features were extracted on CPU: 2 of 4 fields per well, 16 cells per field and the small ViT-S encoder. One of the eight extraction shards did not finish in time, so 5,775 of the 6,600 selected fields are used (every one of the 3,300 wells retains at least one field). The full configuration (ViT-B/14, 4 fields, 32 cells, all fields) is implemented in the same script and is expected to improve the results; we do not claim that here.
- **Viability proxy.** Cell count after 24 h conflates cytostatic and cytotoxic effects and is sensitive to seeding density. At the most toxic doses apoptotic nuclei fragment and are over-counted, occasionally pushing "viability" above 1; Hill fits therefore use a robust (soft-ℓ_1) loss. Dedicated viability stains or LDH release would be better.
- **Conformal guarantee is marginal** and assumes compounds are exchangeable with the reference set; coverage for structurally novel chemotypes may be lower, which the novelty flag partly mitigates.
- **Single-field uploads** are normalised against global rather than plate-matched DMSO statistics and are therefore noisier than plate-level analysis. The demo calibrates its activity percentile on single DMSO fields, reports borderline fields (80th–95th percentile) as low-confidence, and withholds mechanism calls below that; the conformal coverage guarantee applies to treatment-level profiles and is optimistic for a single field. In our bundled examples, a single colchicine field is flagged as borderline and receives a wrong mechanism — exactly the case the low-confidence warning is meant to catch.
- **Segmentation** is classical and tuned for MCF-7 nuclei; dense 3D tissue would need a learned segmenter (e.g. Cellpose).

8 Potential impact

Because Shennong's Glass needs no training, a lab can deploy it on a new chip in a day: image a few dozen reference compounds with known mechanisms and vehicle controls, and every subsequent compound gets an MoA hypothesis with calibrated confidence, a potency/toxicity window and an explanation. Calibrated sets and novelty flags make it safe to use for triage: ambiguous or unprecedented phenotypes are routed to experts instead of silently mislabelled. For neural OoC in particular, the tubulin/actin read-outs at the core of the method are the ones that report axonal and cytoskeletal toxicity, making it a candidate early-warning system for neurotoxic liabilities such as those of microtubule-targeting chemotherapies. We see this as a step towards the data-driven "digital twin" of a chip assay: a reusable, quantitative, uncertainty-aware map from images to

pharmacology.

9 Reproduction

```
pip install -r requirements.txt          # CPU is enough
kaggle auth login                        # free account
bash scripts/fetch_data.sh               # metadata, features, example images
python scripts/run_analysis.py           # results/ + artifacts/  (< 1 min)
python app/app.py                        # http://127.0.0.1:7860
```

Raw-image feature extraction: run `kaggle/features/features.py` (or the sharded CPU variants in `kaggle/cpu_shards/`) as Kaggle notebooks with dataset `stanleyhua/bbbc021` attached, then `scripts/merge_features.py`.

Team

Solo entry: Shreshth Mishra (team leader; AI / computer science). Kaggle: aerochrome.

References

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